

Project Phase I

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Our team will investigate the following big question: **How sensitive are line search methods to the choice of Armijo and Wolfe parameters?** Specifically, we aim to study how variations in the sufficient decrease parameter c_1 and the curvature parameter c_2 influence convergence speed, robustness, and overall computational efficiency. While standard theoretical ranges for these parameters are well known, their practical impact on performance across different optimization algorithms (the algorithms we implement in this project) remains largely empirical. We are particularly interested in understanding whether certain parameter choices consistently improve performance or whether sensitivity depends strongly on the problem structure and algorithm type.

To accomplish this goal, we will conduct a systematic numerical study. For each line search-based algorithm, we will vary c_1 and c_2 and record performance metrics including number of iterations, function evaluations, gradient evaluations, and convergence success rate. We will visualize results using performance profiles, convergence plots (function value vs. iteration) and some other visualization methods. This experimental framework will allow us to quantify parameter sensitivity, identify robust parameter regions, and compare behavior across algorithms. Our findings will guide recommendations for practical parameter selection in line search methods.